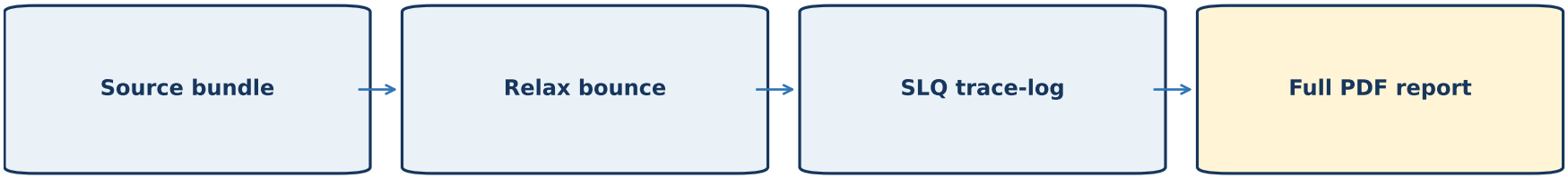
Lattice run report

Independent relax → SLQ jobs | provider-neutral run | numerical results are bare finite-lattice quantities

JOBS PASSED	SITES / SCALAR CHANNEL	PROBE ROWS
1 / 1	810,000	16

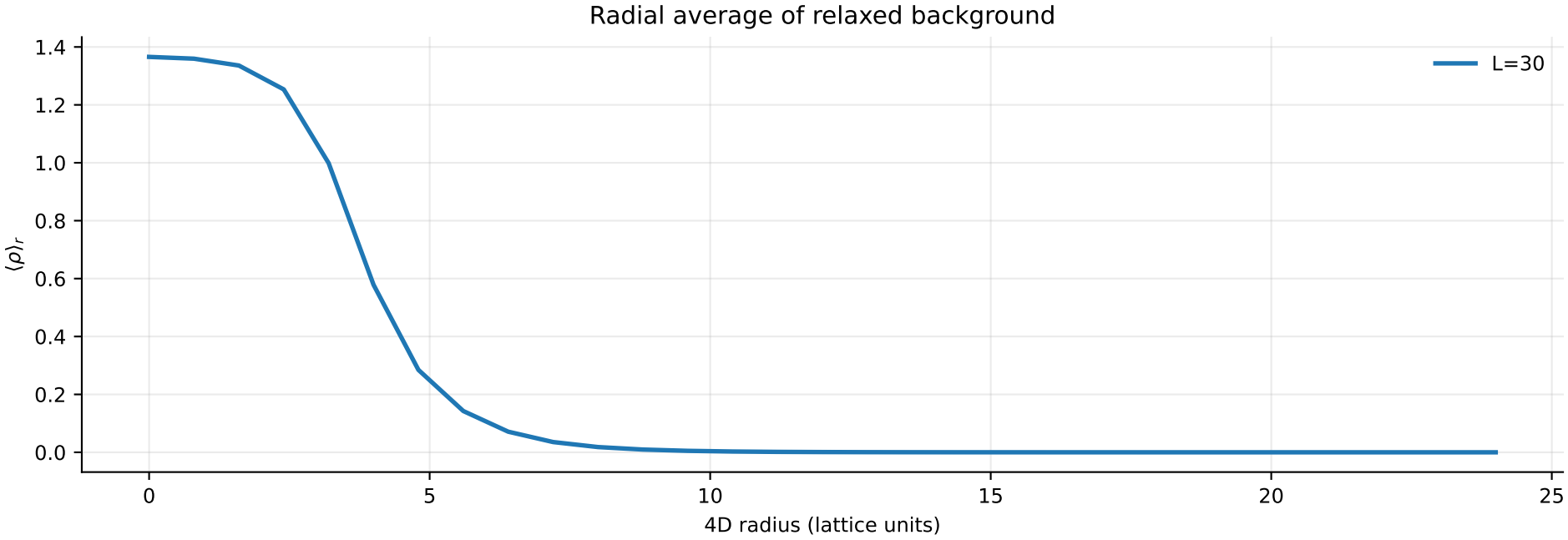


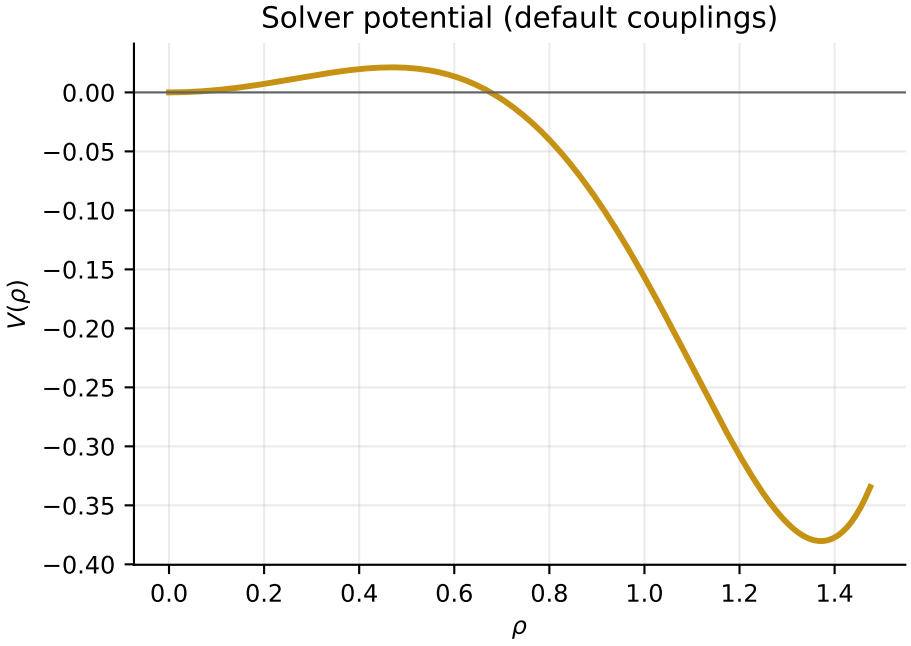
L=30: SUCCEEDED

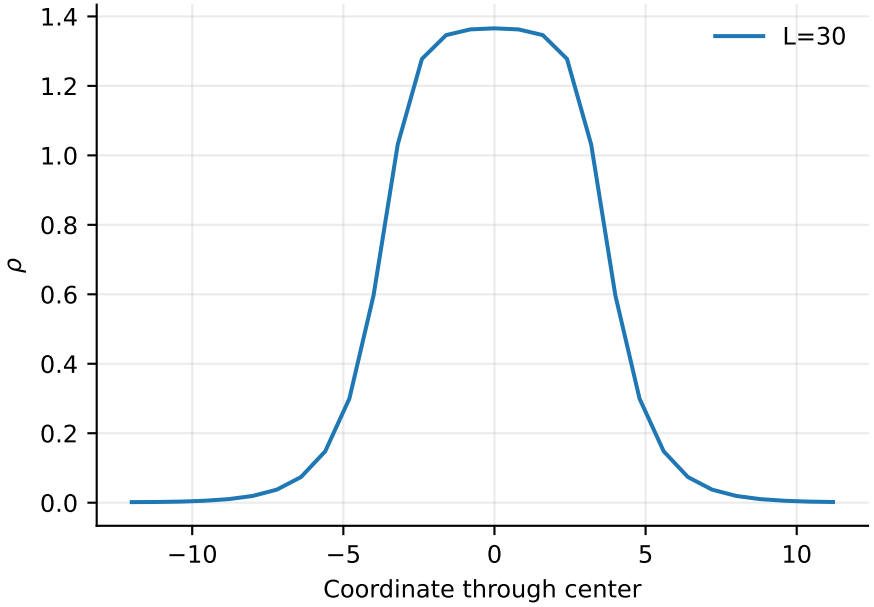
This program runs one lattice size per invocation; relaxation then SLQ run sequentially.
The source archive records the exact runner, solver, and nonlinear SU(2) vertex module.

Relaxed field and potential

Spatial diagnostics reconstructed from the saved relaxed-field checkpoints

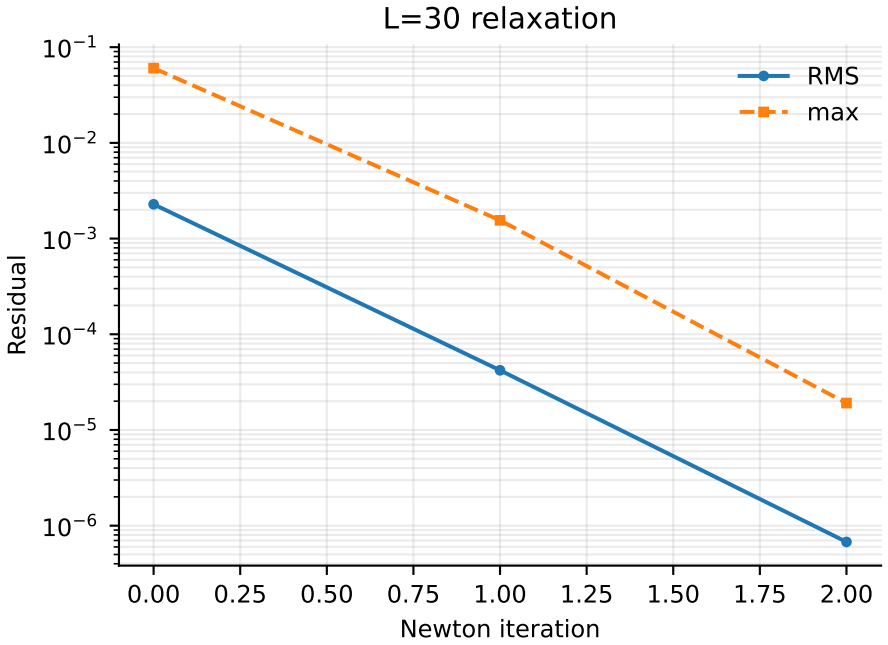


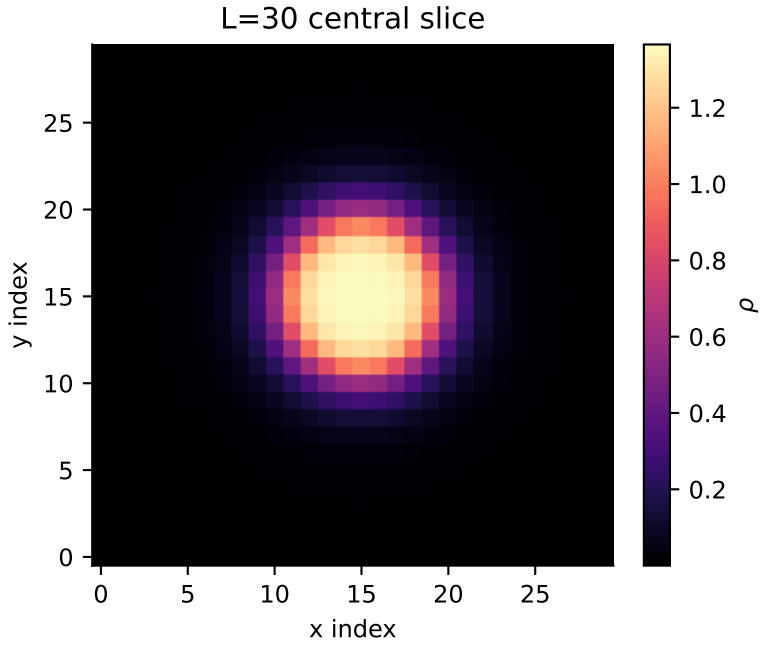




Relaxation convergence and field slices

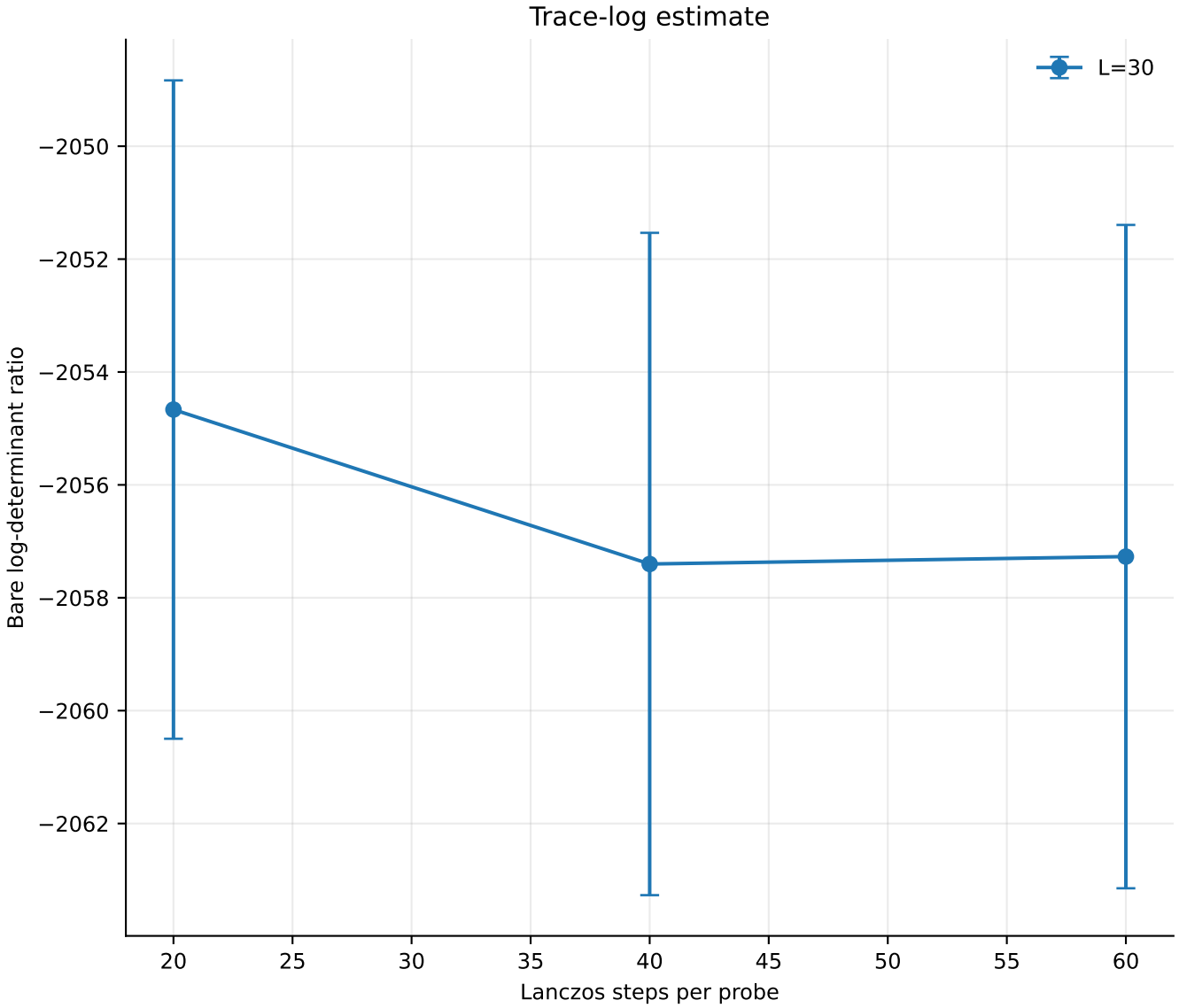
Newton history and central x–y slices at the midpoint of the other coordinates

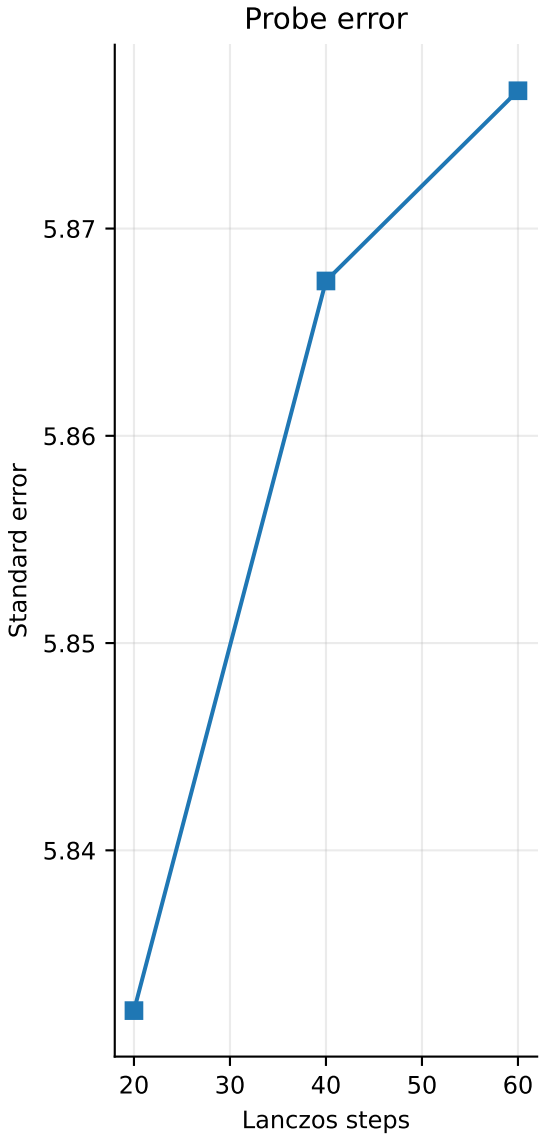




SLQ convergence by Lanczos order

Means and probe standard errors from slq_summary_by_steps.csv

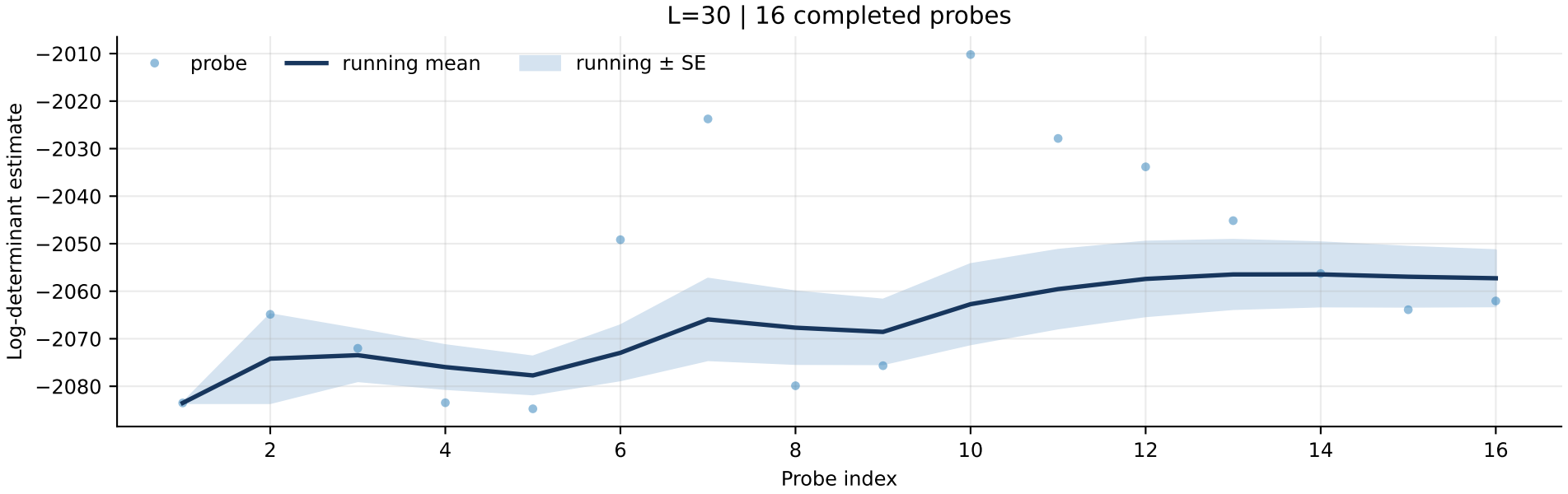


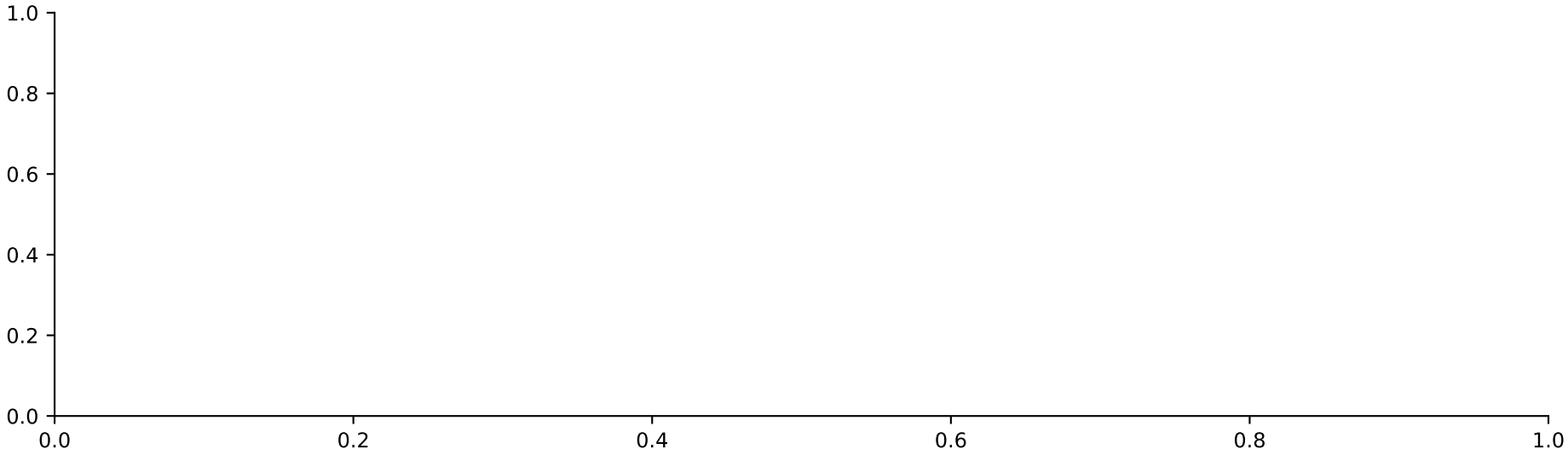


Interpretation: these are bare finite-lattice SLQ estimates. The plotted errors are stochastic probe errors; they do not include discretization, continuum matching, or regulator/extrapolation uncertainties.

Probe ensemble and stochastic stability

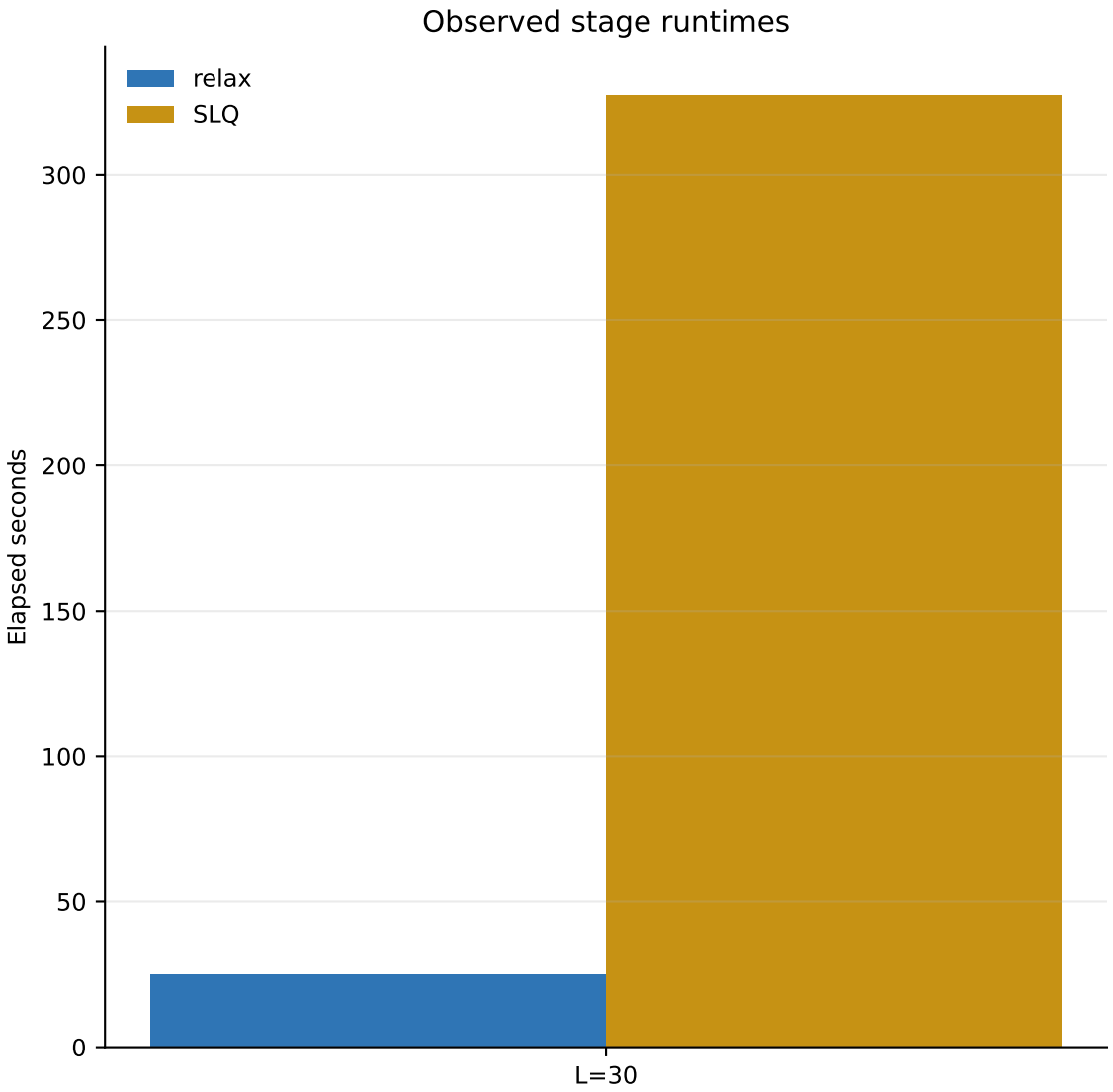
Per-probe estimates at the largest requested Lanczos order

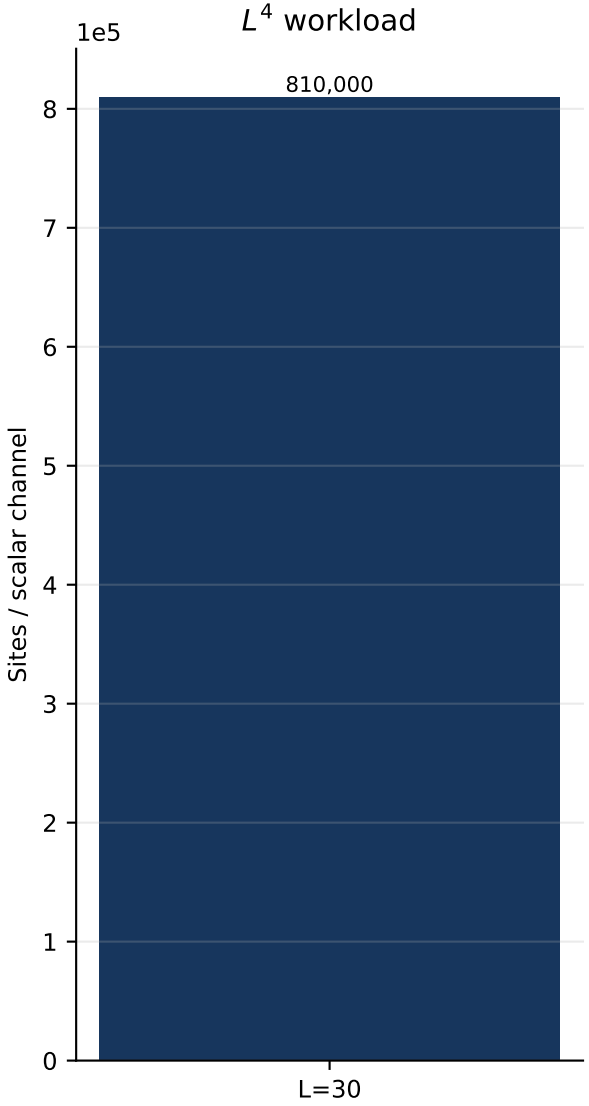




Execution profile and lattice scaling

Measured stage wall times only; no cloud price or peak-memory estimate is inferred





This invocation processes one lattice size at a time with one solver process.
Wall time and memory depend on the selected host and numerical-library build.

Results ledger and reproducibility record

A compact handoff page for review, archiving, and cloud-worker transfer

Job	Overall	Final RMS	Bare logdet ± SE	Probes	Relax / SLQ
L30_h0.8	SUCCEEDED	3.273e-12	-2057.27 ± 5.9	16 / 16	succeeded/succeeded

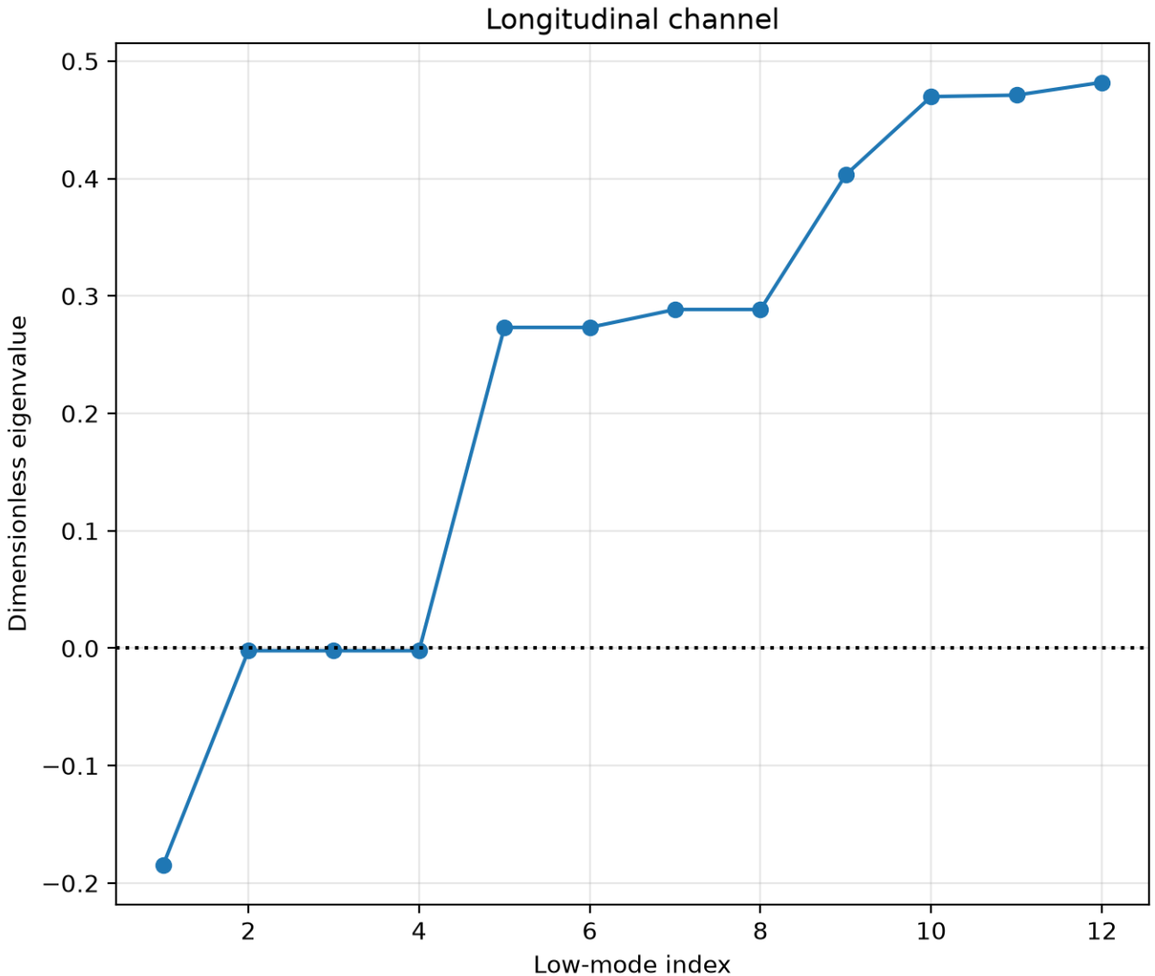
Spacing h=0.8 | probes/job=16 | Lanczos steps=[20, 40, 60] | workers=1 | threads/worker=1

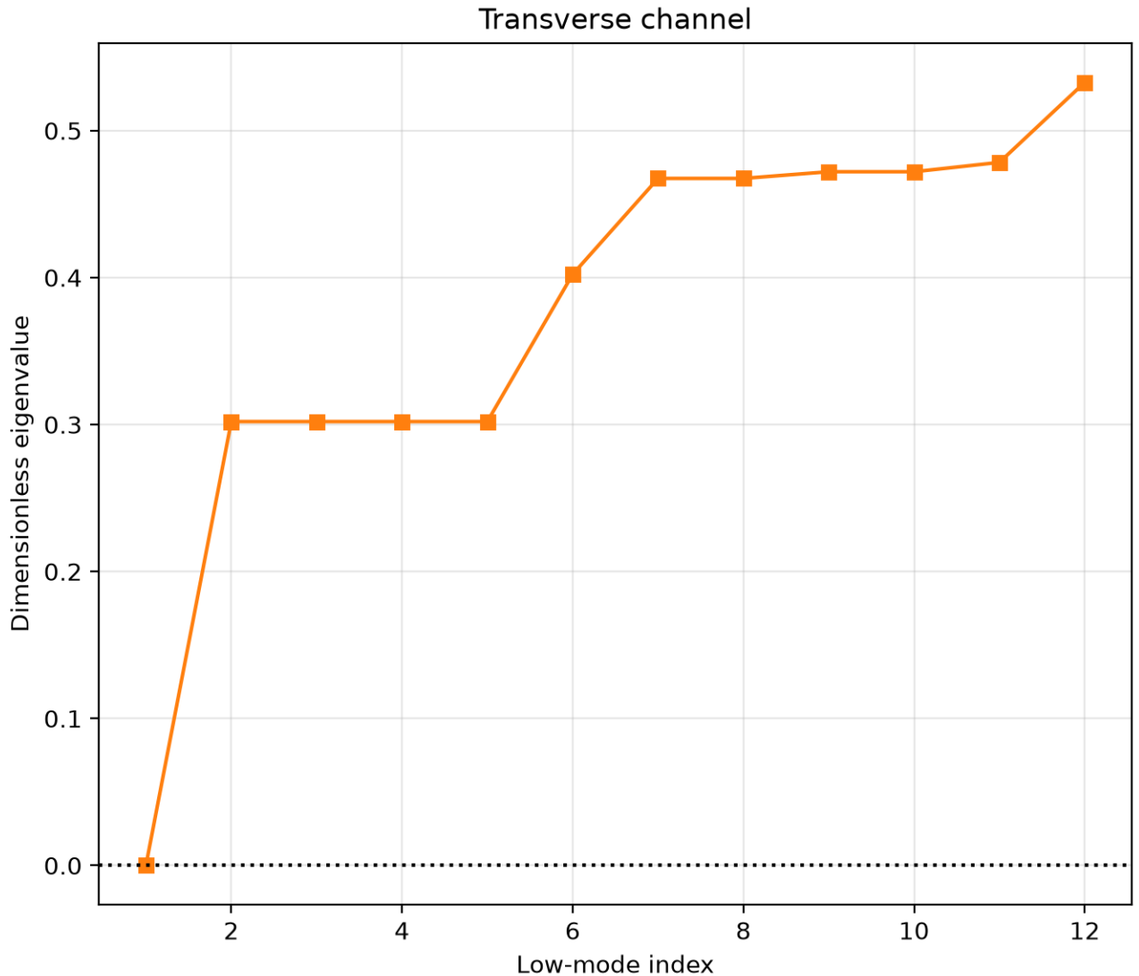
Source archive: campaign_source_bundle.zip | SHA-256: 545316e28d7979c4b648c981ca1b8fac53a66301af45648918b4f3a1975ac40d

Scope: one scalar channel lattice count is shown; the determinant is bare and finite-lattice. This report does not establish continuum matching, spacing extrapolation, a gauged decay-rate prefactor, cloud billing, or peak RAM. Uniform stationary fields are not localized bounces. Preserve the manifests and

Deflated Low Modes Used by the SLQ Trace-Log

$L = 30$, $h = 0.8$, sites/channel $= 810000$, probes $= 16$, deflated modes/channel $= 12$

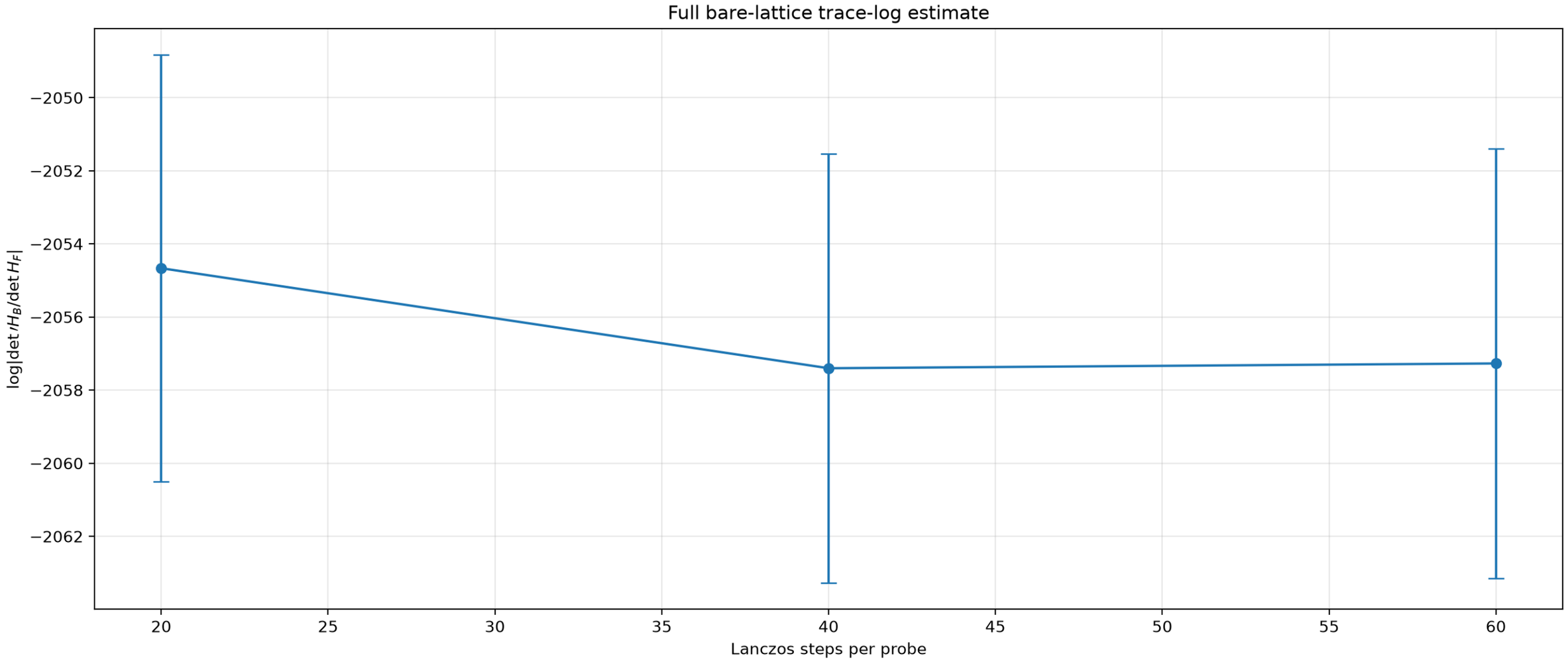




The radial negative mode is retained through log|lambda|. Four translation modes and three internal-orientation modes are removed from the primed determinant.

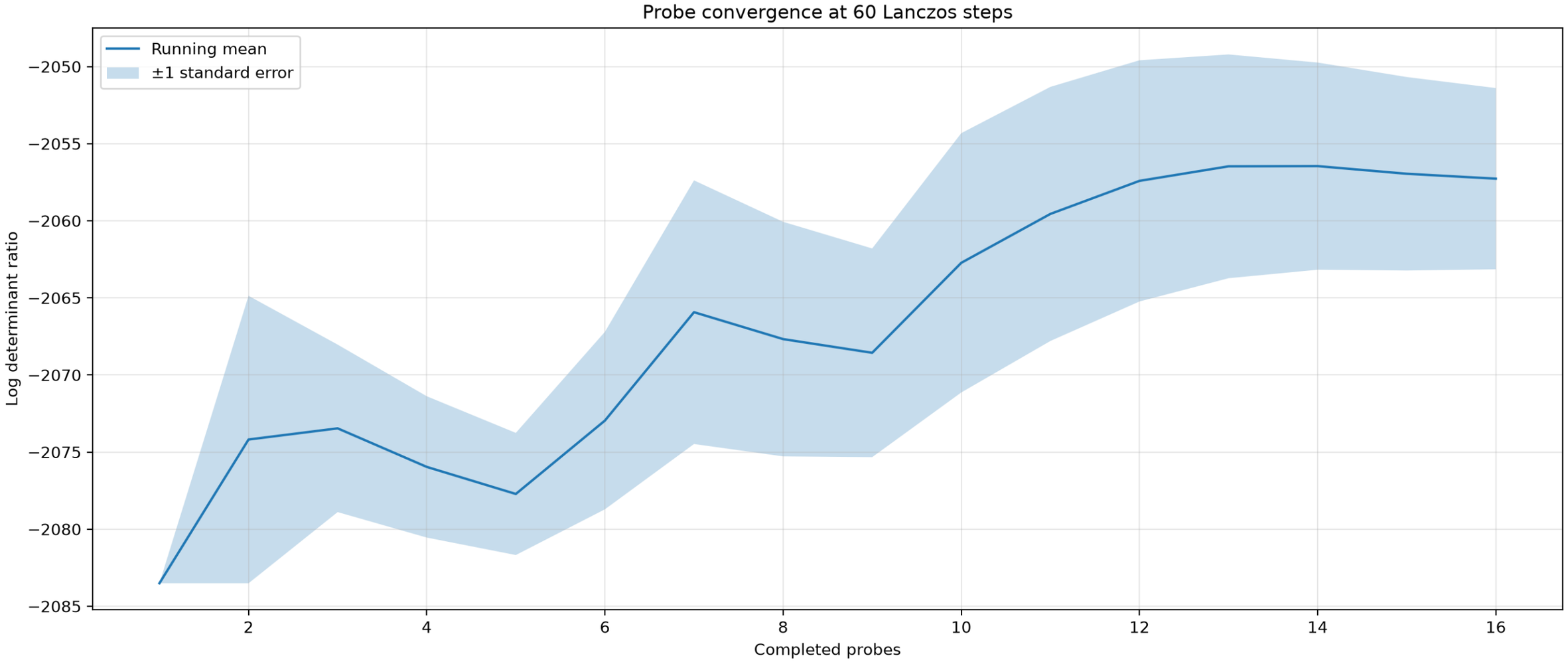
SLQ Lanczos-Order Convergence

$L = 30$, $h = 0.8$, sites/channel $= 810000$, probes $= 16$, deflated modes/channel $= 12$



SLQ Stochastic-Probe Convergence

$L = 30$, $h = 0.8$, sites/channel $= 810000$, probes $= 16$, deflated modes/channel $= 12$



Relaxed-Bounce Full Lattice Trace-Log Status

$L = 30$, $h = 0.8$, sites/channel $= 810000$, probes $= 16$, deflated modes/channel $= 12$

Quantity	Value	Status
Geometry	30^4, h=0.8	810000 sites/channel
Physical negative mode	-1.84992623e-01	retained
Translation modes	4	removed
Orientation modes	3	removed
Final bare log determinant ratio	-2.05727086e+03	+/- 5.877e+00
Log bare determinant factor	1.02863543e+03	+/- 2.938e+00
Base-10 factor exponent	446.730691	+/- 1.276
Probe checkpoint	probe_checkpoint.csv	resumable

This completes the bare finite-lattice trace-log. Continuum matching and an h->0 extrapolation remain required for a renormalized physical determinant.